

KAHYUN KIM

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EDUCATION

Korea University, Seoul, South Korea
B.S. in Statistics

Mar 2024–Present
GPA: 4.21 / 4.50

University of British Columbia, Vancouver, Canada
Exchange Student, Faculty of Science (Statistics)

Jan–Apr 2026
Term Average: 88.3%

RESEARCH EXPERIENCE

Propofol Control with PPO and VitalDB, Korea University
Undergraduate Research Intern, Prof. Joongheon Kim's Lab

2026

- Designed a 2-by-2 study of causal BIS visibility and state representation for PPO-based propofol control in a reconstructed PK–PD simulator; evaluated 490 sealed-test cases from 483 subjects.
- Estimated a 6.64-point preprocessing–state interaction (95% CI: 6.29–6.98) in subject-level mean absolute BIS deviation; assessed BIS-range time and propofol exposure.

Virtual Cell Challenge 2026

Sep 2026–Present

KUBIG Team Conference (Korea University Data Science & AI Society)

- Diagnosed effect-size shrinkage: 80.7% of 30,262 false negatives had the correct direction but fell below the significance threshold.
- Developed a calibrated low-dose Soft-U loss; improved all six compact-model metrics on 30 unseen targets, including direction reach from 0.245 to 0.282. Full-scale evaluation is ongoing.

PUBLICATION

Kahyun Kim, Hyojun Ahn, Hyung-Chul Lee, and Joongheon Kim. “Joint Observation Preprocessing and Feature-Group Ablation for State Representation Optimization in Reinforcement Learning-Based Propofol Control.” *International Conference on Information and Communication Technology Convergence (ICTC)*, 2026. Accepted.

SELECTED PROJECTS

Korean Financial News RAG, KUBIG (four-person team)

Aug 2026

- Built BGE-M3/Chroma indexing and designed a 50-question paired retrieval evaluation; hybrid Hit@1 was 0.90 vs. 0.78 dense (paired $p = 0.070$).

Canadian Influenza Positivity Forecasting

Jun 2026

- Modeled 2015–2019 WHO FluNet weekly data with ARIMA and SARIMA; reduced 52-week hold-out RMSE from 8.77 to 2.20 with a seasonal model.

Speech Emotion Classification, OUTTA AI Bootcamp

Aug 2025

- Built a six-class speech emotion CNN with MFCC and attention pooling (56.7% validation accuracy); received the Outstanding Participant Award.

TECHNICAL SKILLS

Programming & Tools: Python, R, SQL, PyTorch, Git

Statistical Methods: ARIMA/SARIMA forecasting, STL decomposition, nonparametric tests, TOST, tolerance intervals, clustering, classification

Languages: Korean (native), English (IELTS Academic 8.0)